Autonomous AI Agent State Machine for Customs Clearance

Supply Chain Research — System Design Specification

Version 3.0 | Customs Clearance Only | Data-Aware LLD | Human-in-the-Loop

研究目标： 构建一个可以在海关清关业务中自主观察、推理、规划、调用工具、验证结果并恢复异常的 AI Agent。自主性仅限于获得授权的业务操作；海关放行和监管决定必须由可信外部事件确认。

核心原则： AI 自主规划与协调；State Machine 跟踪执行；Policy Engine 约束权限；外部监管与授权系统提供权威业务事实。

V3.0 Scope and Research Boundary

In scope: inbound shipment/entry data, commercial documentation, valuation evidence, tariff classification research, country of origin, entry rejection, CBP hold/inspection coordination, partner-government-agency (PGA) review, duty/bond evidence, official CBP/PGA status monitoring, exception recovery, and role-specific approvals. All regulatory activities remain subject to applicable authorization.

Out of scope: terminal pickup, drayage, warehouse distribution, last-mile delivery, and any direct production ACE filing without an authorized interface and responsible legal actor. `CLEARANCE_VERIFIED` represents the experiment's final business condition: trusted evidence indicates applicable CBP and PGA clearances and any task-specific clearance prerequisites have been satisfied; it is *not* an AI-issued customs release.

Method: official documentation informs the fields and workflow; synthetic transaction records and simulated messages supply research tasks. No simulated case is asserted to be an actual CBP adjudication, official status code, or real shipment record.

1 Two-Layer State Model

Agent State： AI 当前执行的认知或操作阶段，例如 `OBSERVE`, `REASON`, `PLAN`, `ACT`, `VERIFY`, `RECOVER`。

Shipment State： 货物真实的清关业务阶段，例如 `ENTRY_SUBMITTED`, `CUSTOMS_HOLD`, `CBP_RELEASED`, `CLEARANCE_VERIFIED`。

二者必须独立存储。例如，当货物仍处于 `CUSTOMS_HOLD` 时，Agent 可以经过多轮观察、执行与恢复；文件核对成功不能直接改变海关放行状态。

2 Execution Ownership and Human Roles

Execution modes are assigned per action, not just per agent state. The same `ACT` state may run a read-only AI tool, generate an AI-assisted filing draft, or wait for a legally authorized actor to submit a customs entry.

Therefore `execution_mode`, `responsible_role`, `approval_policy`, and `authorized_executor` must be attached to each action.

Mode	Control ownership	Meaning / 边界
AI	Autonomous agent	在预授权范围内自动分析、读取、请求和执行低风险工具。
HYBRID	AI proposal + named human	AI 给出证据与建议；指定角色审核或提供事实确认后才继续。
HUMAN / AUTHORITY	Authorized person or regulator	人类/有权机构掌握最终决定或执行权；AI 不得替代官方状态。

2.1 Stakeholder and Role Catalog

表 1: Human and external authority roles

ID	Human / Organization	Responsibilities and authority
R1	Supply Chain Operator	跟踪任务、处理运营异常、协调物流；不自动拥有申报审批权。
R2	Warehouse Worker / Supervisor	现场核实实物数量、包装、入出库记录与装卸异常。
R3	Factory / Supplier Representative	出具或确认 Invoice、Packing List 和生产/商业资料；通常为外部用户。
R4	Licensed Customs Broker	在合法授权范围内处理清关申报、更正、与海关沟通等专业报关业务。
R5	Importer / Import Compliance Officer	对进口商信息、分类依据、合规风险和内部审批负责。
R6	Customs Officer / CBP / Relevant Agency	外部监管权威；负责其法定管辖下的 Hold、查验、Release 等决定；不是企业内部审批用户。
R7	System Administrator	维护权限、密钥、运行平台和技术故障；无默认清关业务批准权。
R8	Logistics / Transportation Coordinator	判断运输可行性、安排卡车及已授权的提货交接。
R9	Finance / Duty Payment Officer	审核账单、税费资金安排和内部付款授权。

Role vs. identity: a role specifies what a party can do; identity and permissions specify which real account may do it for a particular shipment. CBP is modeled as an *external authority/event source*, not as a user assigned a task in our internal approval queue.

2.2 Human–AI Responsibility Matrix

表 2: Customs tasks, execution mode and accountable actor

Business Task	Mode	AI Responsibility	Human / Authority
Shipment status collection	AI	Query ERP/TMS, validate provenance	R1 if disputed
Document OCR and extraction	AI	Extract and normalize fields	R3 for source corrections
Document consistency check	AI	Compare Invoice / Packing List / PO	R2/R3 when discrepancy found
Missing document resolution	HYBRID	Identify gaps, request and recheck	R3; R2 if stock-related
Quantity discrepancy	HYBRID	Compare records, produce evidence	R2 + R3
HTS classification research	HYBRID	Suggest candidates and rationale	R4 / R5
Entry draft preparation	HYBRID	Validate data; prepare draft	R4 / authorized importer
Customs filing or amendment	HUMAN	Present approved data and status	Authorized R4 / importer
CBP hold investigation	HYBRID	Analyze notices; propose plan	R4 + R5; R6 decides official status
Physical customs examination	AUTHORITY	Monitor trusted notifications	R6 / authorized inspection personnel
Regulatory release	AUTHORITY	Observe and validate official notice	R6 / relevant agency
Duty/payment review	HYBRID	Estimate or reconcile figures	R5 / R9
Post-clearance pickup	OUT OF SCOPE	Not executed in this study	Future work
Technical incident recovery	AI/HYBRID	Safe retries, event reconciliation	R7 when privileged changes required

Authority invariant: recommendations, drafts, or successful tool calls are not customs authorization. No AI-generated message may set `cbp_released=true`; that flag requires an authenticated official/authorized release event.

3 Autonomous Agent State Transition Diagram

Figure 1 separates the main execution cycle from exceptional paths. To keep the figure legible, only major transitions are drawn; all event-driven return paths are specified precisely in Table 4.

4 Human-in-the-Loop Role Routing

A generic `ESCALATE` node is insufficient. The **Human Role Router** must classify the approval purpose, find an authorized principal, persist the proposed action version, and wait for an authenticated decision. For all reg-

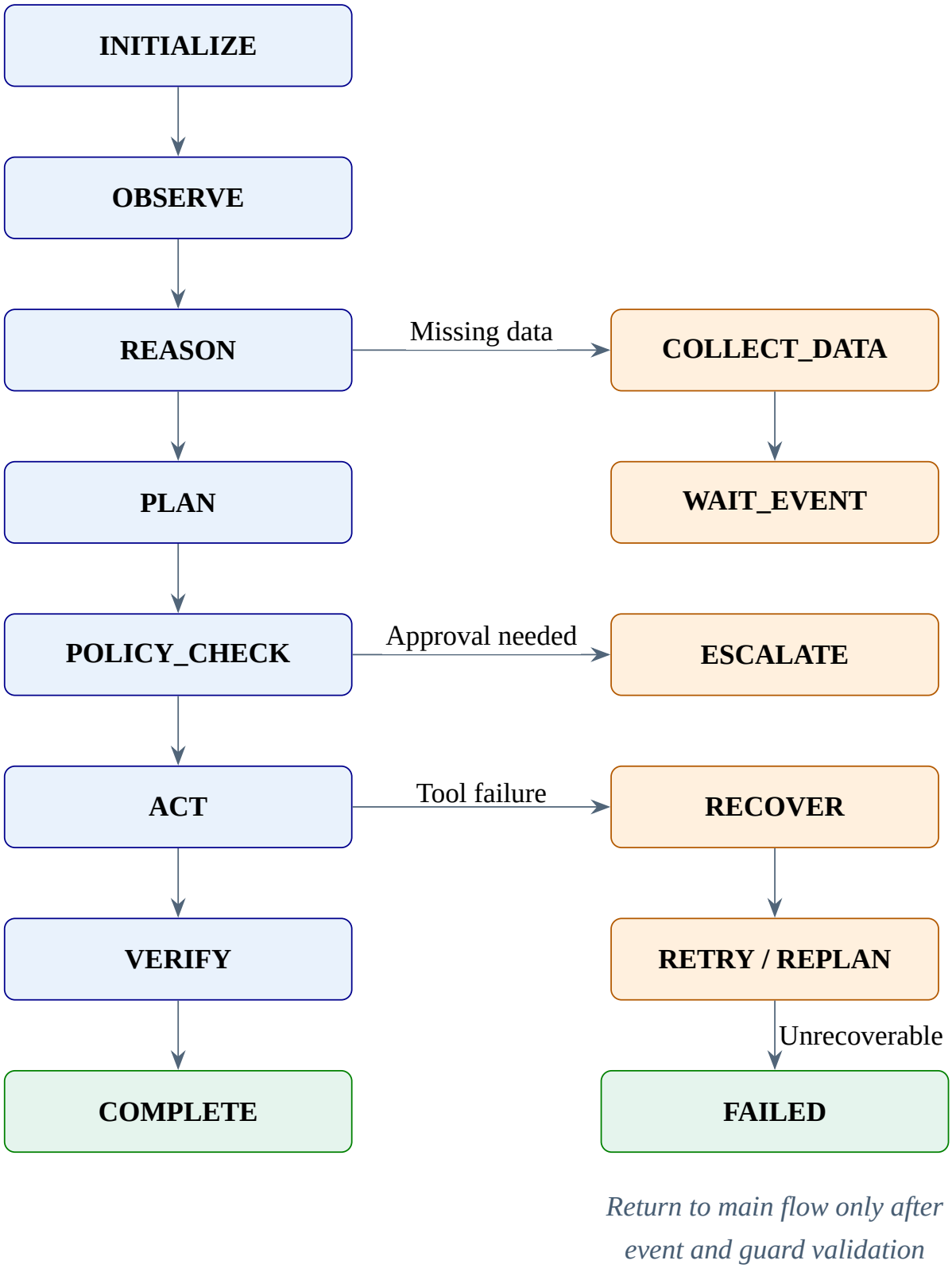


图 1: Clean overview of the autonomous agent execution cycle and exception-handling lane. The diagram intentionally omits looping arrows to prevent crossings; missing-data events resume at OBSERVE, approved actions resume through POLICY_CHECK, and recovery actions return to ACT or PLAN only after validation. Exact transitions are in Table 4.

ulatory release decisions, the workflow waits for an external authoritative event rather than assigning approval to an internal user.

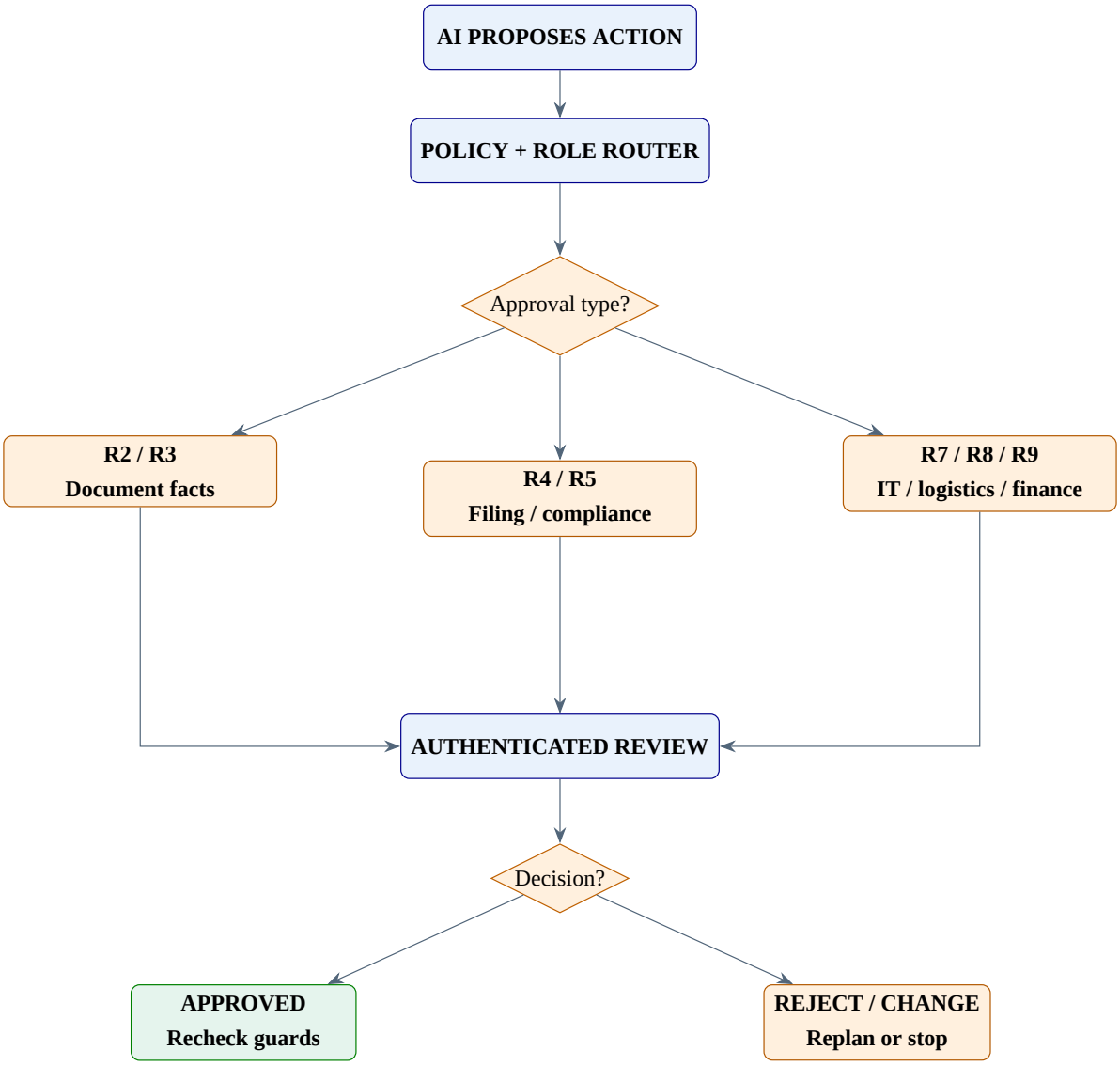


图 2: Role-aware human-in-the-loop routing. The router resolves company-side human reviewers by action category; customs authorities are handled through external official events, not this internal approval queue.

4.1 Role Selection Policy

Trigger / Issue	Target and decision rights
Packing List mismatch	R3 confirms supplier source; R2 independently checks physical quantity.
Classification ambiguity / entry amendment	R4 or an otherwise authorized importer handles professional filing; R5 owns applicable internal import-compliance review.
CBP Hold / examination	R4 coordinates and R5 monitors impact; R6 or official agency channel supplies authoritative outcome.
Access token or connector outage	R7 handles privileged infrastructure changes, never substantive customs approval.
Pickup or trucking conflict	R8 confirms schedule and transportation authorization.
Duty-payment exception	R9 confirms payment; R5 reviews compliance significance if needed.

5 Agent States and Responsibilities

State ownership is not identical to execution ownership. `ACT`, `COLLECT_DATA`, `VERIFY`, `RECOVER`, and `ESCALATE` can operate in different execution modes depending on the pending action. A human review is always attributed to an explicit role and user identity; `POLICY_CHECK` is a deterministic guard rather than an LLM self-approval.

State	Responsibility / 职责	Output
INITIALIZE	验证任务、Shipment ID、目标和访问权限	Task Context
OBSERVE	获取运输、文件、报关以及历史事件	Observation
REASON	分析异常、原因及不确定性	Analysis
COLLECT_DATA	主动获取缺失文件或关键字段	Data Request
PLAN	将任务拆分为多步可执行计划	Action Plan
POLICY_CHECK	检查权限、合规限制和审批要求	Policy Decision
ACT	调用获授权的工具并记录执行结果	Tool Result
VERIFY	验证行动效果及总体任务目标	Verification
WAIT_EVENT	持久化任务，等待异步消息唤醒	Pending Event
RETRY	执行安全、有限次数的重试	Retry Request
REPLAN	在原计划失败时制定可行替代方案	Revised Plan
RECOVER	分类错误，核实外部状态，选择恢复策略	Recovery Decision
ESCALATE	请求人工判断、批准或处理	Approval Request
COMPLETE / FAILED	记录任务完成或终止证据	Final Report

6 Shipment Business States

业务路径（研究抽象，而不是 CBP 官方状态代码）：

$$\texttt{PRE_ARRIVAL} \rightarrow \texttt{DOCUMENT_PREPARATION} \rightarrow \texttt{ENTRY_READY}$$
$$\rightarrow \texttt{ENTRY_SUBMITTED} \rightarrow \texttt{CUSTOMS_PROCESSING}$$
$$\rightarrow \texttt{CBP_RELEASED} \rightarrow \texttt{CLEARANCE_VERIFIED} \rightarrow \texttt{DELIVERED}$$

异常分支：

$\texttt{CUSTOMS_PROCESSING} \rightarrow \texttt{CUSTOMS_HOLD} \rightarrow \texttt{CUSTOMS_PROCESSING}$

$\texttt{ENTRY_SUBMITTED} \rightarrow \texttt{ENTRY_CORRECTION_REQUIRED} \rightarrow \texttt{ENTRY_READY}$。

实际清关申报可以发生在到港前，税费、其他监管机构（PGA）和运输放货要求可能并行。因此不能仅靠一个业务状态标识推断货物已可提取。

7 State Transition Specification

定义 $T = (S, E, G, A, S')$，分别表示当前状态、事件、守卫条件、行动和下一状态。

表 4: Agent State Transition Table

Current State	Event	Guard	Next State
INITIALIZE	TASK_CREATED	G1	OBSERVE
OBSERVE	DATA_RECEIVED	G2	REASON
REASON	INSUFFICIENT_DATA	G3	COLLECT_DATA
REASON	READY_TO_PLAN	G4	PLAN
COLLECT_DATA	REQUEST_SENT	G5	WAIT_EVENT
PLAN	PLAN_GENERATED	G6	POLICY_CHECK
POLICY_CHECK	ACTION_ALLOWED	G7	ACT
POLICY_CHECK	APPROVAL_REQUIRED	G8	ESCALATE
POLICY_CHECK	ACTION_DENIED	G9	REPLAN
ACT	TOOL_SUCCEEDED	G10	VERIFY
ACT	TOOL_PENDING	G11	WAIT_EVENT
ACT	TOOL_ERROR	G12	RECOVER
VERIFY	GOAL_COMPLETED	G13	COMPLETE
VERIFY	TASK_REMAINS	G14	OBSERVE
VERIFY	RESULT_INVALID	G15	RECOVER
WAIT_EVENT	NEW_EVENT	G16	OBSERVE
WAIT_EVENT	TIMEOUT	G17	RECOVER
RECOVER	RETRY_SELECTED	G18	RETRY
RECOVER	REPLAN_SELECTED	G19	REPLAN

Current State	Event	Guard	Next State
RECOVER	ESCALATION_NEEDED	G20	ESCALATE
RECOVER	UNRECOVERABLE	G21	FAILED
RETRY	RETRY_READY	G22	ACT
REPLAN	NEW_PLAN_GENERATED	G6	POLICY_CHECK
ESCALATE	HUMAN_APPROVED	G23	POLICY_CHECK
ESCALATE	HUMAN_REJECTED	G24	REPLAN / FAILED

8 Guard Conditions (G1–G24)

Guard	Deterministic Condition
G1	Task ID、Shipment ID、目标有效且任务可接收。
G2	数据来源可信、字段格式有效。
G3	证据不足，存在可收集的附加数据。
G4	分析结果能支持明确、可执行的计划。
G5	资料请求已被系统接受并保存。
G6	计划结构正确，工具、参数与依赖关系有效。
G7	操作在授权范围内且全部合规限制满足。
G8	操作需要人工授权或专业判断。
G9	操作被策略引擎明确禁止。
G10	已获得可验证的工具结果。
G11	操作已受理但尚未产生最终结果。
G12	已捕获执行错误或外部结果未知。
G13	所有任务完成条件已有独立证据验证。
G14	当前操作成功但总体目标尚未完成。
G15	操作结果未通过后置条件检查。
G16	新事件通过来源认证、关联匹配和去重。
G17	到达预设截止时间，且未获得期望事件。
G18	故障为可重试类型，并满足安全前提。
G19	当前计划失败且存在安全替代方案。
G20	存在重大风险、缺乏授权或证据不足。
G21	无法通过安全且合法的方式继续任务。
G22	重试次数未超限、外部结果已核实、幂等保护有效。
G23	授权人、审批范围、操作内容和有效期均匹配。
G24	审批明确拒绝或已失效。

Global Guard：严重合规阻断事件应暂停相关行动，进入 `ESCALATE` 或 `WAIT_EVENT`。文件异常修复不等于海关 Hold 已解除；只有经授权的官方事件才能更新 `cbp_released=true`。

8.1 Additional Role and Approval Guards (G25–G28)

Guard	Deterministic Condition
G25	**ROLE_RESOLVED**: required reviewer role maps to a specific eligible principal or a declared external authority channel.
G26	**APPROVER_AUTHORIZED**: approver is authenticated and has shipment/action-scoped permission; technical admin role alone is insufficient.
G27	**APPROVAL_VALID**: approval is within its validity period and binds the action ID, payload hash, evidence snapshot and version. Material changes invalidate approval.
G28	**EXECUTOR_AUTHORIZED**: acting principal is allowed to perform the final operation under organization policy and applicable legal requirements.

New events: `ROLE_RESOLVED`, `APPROVAL_REQUESTED`, `APPROVAL_GRANTED`, `APPROVAL_REJECTED`, `APPROVAL_EXPIRED`, `ACTION_VERSION_CHANGED`, `AUTHORITY_EVENT_RECEIVED`.

From `ESCALATE`, an approved action must satisfy G25–G28 and repeat `POLICY_CHECK` before `ACT`. Rejected or expired approvals do not authorize execution.

9 Event Taxonomy

Category	Example Events	Origin
Shipment	`CARGO_ARRIVED`, `ETA_CHANGED`	Carrier / TMS
Document	`INVOICE_RECEIVED`, `DOCUMENT_MISMATCH`	ERP / Document Agent
Customs	`ENTRY_REJECTED`, `CBP_HOLD_RECEIVED`, `CBP_RELEASE_RECEIVED`	Authorized Channel
Agent	`TOOL_FAILED`, `PLAN_COMPLETED`, `RETRY_EXHAUSTED`	Agent Runtime
Human	`APPROVAL_GRANTED`, `DOCUMENT_PROVIDED`	Approval System

10 Agent Tool Contract

Tool	Function	Permission
`get_shipment_status()`	查询物流与货物状态	Read
`get_customs_status()`	查询可信清关状态	Read
`read_documents()`	提取文件信息	Read
`compare_documents()`	检查发票、装箱单差异	Internal
`check_compliance_rules()`	辅助查找监管要求	Advisory
`request_missing_documents()`	向供应商请求文件	Authorized Write
`create_broker_task()`	创建报关行协作任务	Authorized Write
`request_human_approval()`	提交人工审批请求	Approval
`submit_customs_entry()`	由合法授权的主体完成申报	Restricted
`schedule_pickup()`	满足条件时安排提货	Conditional Write

11 Exception Recovery

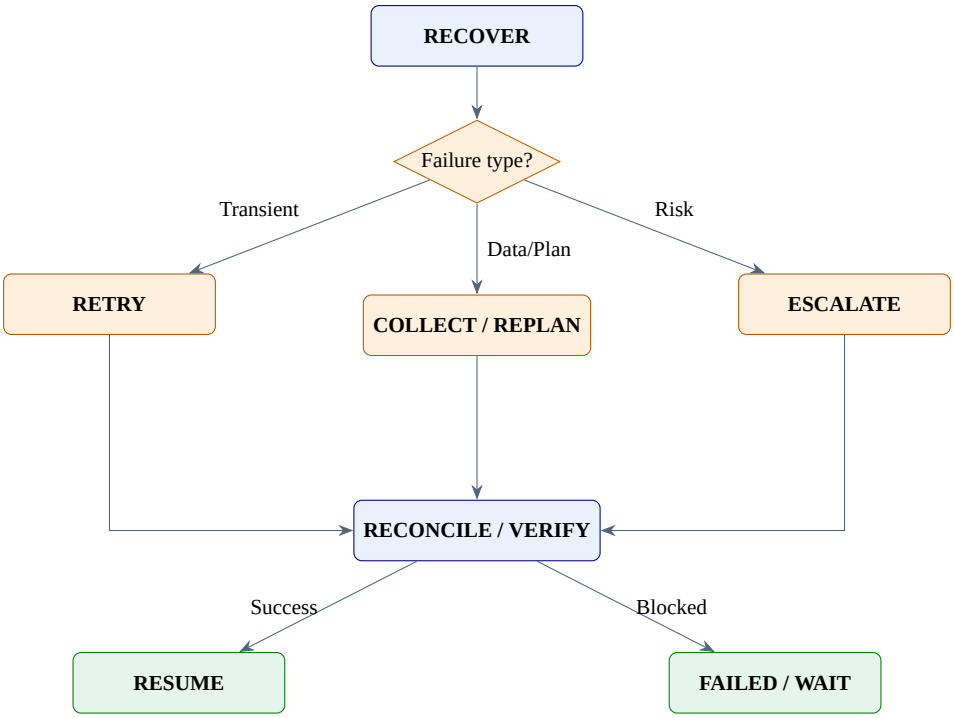


图 3: Recovery Policy：按异常类型选择恢复操作，恢复后再次核实实际结果。

Exception	Example	Strategy
TRANSIENT_FAILURE	API 503 / 查询失败	有上限的指数退避重试
DATA_MISSING	缺少 Invoice	请求补件并等待
DATA_CONFLICT	数量不一致	收集证据、请求确认
PLAN_FAILURE	行动无法推进	重新规划
REGULATORY_BLOCK	CBP Hold	联系授权主体并等待官方处理
UNKNOWN_OUTCOME	提交后网络超时	查询外部状态，禁止盲目重发

Recovery Invariants：

- 对可能产生副作用的工具操作使用幂等键与审计记录。
- 结果未知时先查询外部状态，避免重复提交。
- 为重试次数、等待时间、任务预算设置上限。
- 每次恢复前后保存事件、证据与状态快照。
- AI 不能自行解除官方 Hold 或伪造 Release。

12 Case Study: Document Mismatch and Customs Hold

Step	Agent Action and Expected Evidence
1 OBSERVE	收到可信 Hold 消息；Invoice=500 units，Packing List=480 units。
2 REASON	发现文件差异，但不假定其一定是官方 Hold 原因。
3 PLAN	制定仓库数量核查、供应商补件和报关行协调计划。
4 POLICY_CHECK	核实文件请求与报关行协作工具的授权范围。
5 ACT	发送补件请求、创建工单，保存外部请求 ID。
6 WAIT_EVENT	等待供应商提供新文件，并响应可信的文件更新事件。
7 VERIFY	比对修订文件、原始订单及盘点证据，仅关闭文件异常。
8 WAIT_EVENT	等待报关行与海关进一步处理消息。
9 VERIFY	根据可信来源核实 CBP、PGA 及适用放货要求。
10 COMPLETE	在完成条件全部成立时移交后续运输任务。

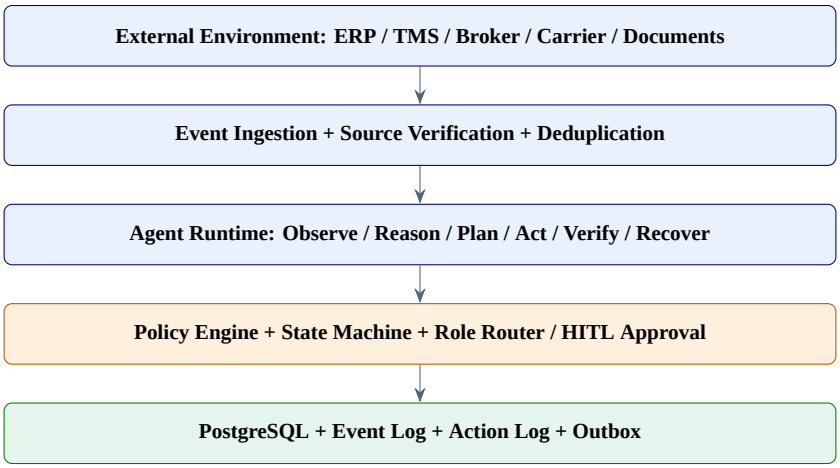
13 Role-Aware Case: Document Mismatch + Customs Hold

For an Invoice with 500 units and a Packing List with 480 units, the AI may detect the mismatch and ask for verification, but cannot infer that the discrepancy caused the official hold or change customs release status. The following sequence identifies the *responsible actor* explicitly.

Stage	Mode	Actor	Action and evidence
Observe	AI	Agent	Read trusted Hold event and two documents; detect 500 vs. 480.

Stage	Mode	Actor	Action and evidence
Investigate	HYBRID	AI + R2 / R3	Request packing and stock confirmations; retain replies as evidence.
Verify documents	AI/HYBRID	Agent + R2/R3	Reconcile corrected list against stock and commercial records.
Review customs action	HYBRID	AI + R4 / R5	Prepare proposed remedy, obtain appropriate approval for filing changes.
File / coordinate	HUMAN	R4 / authorized importer	Conduct authorized customs business; Agent tracks acknowledgement.
Official decision	AUTHORITY	R6 / official channel	Hold/release determined externally and recorded with signed/trusted evidence.
Transportation	AI/HYBRID	AI + R8	After all release conditions are verified, schedule or approve pickup.

14 System Architecture and Data Design



Recommended tables:

- `agent_tasks`, `agent_events`, `agent_actions`
- `approvals`, `roles`, `role_assignments`

Store `task_id`, `shipment_id`, `goal`, `agent_state`, `shipment_state`, `current_plan`, `retry_count`, `awaiting_event`, and `version`.

Approval record fields:

- `action_id`, `execution_mode`, `required_role`, `approver_id`
- `authorized_executor`, `approval_status`, `action_version`
- `payload_hash`, `evidence_snapshot_id`, `expires_at`

Any material change to an approved filing draft invalidates the previous approval.

每条状态转换应记录 `event_id`, `event_source`, `timestamp`, `guard_result`, `action_result` 与证据引

用。对并发更新可使用版本号检查及事务性 Outbox Pattern。

15 Research Evaluation and MVP

System	Planning	Error Handling
Baseline 1: Rule-Based FSM	固定步骤	预设异常分支
Baseline 2: LLM-Only Agent	LLM 自主规划	主要依赖 LLM 推理
Proposed Agent FSM	自主规划 + 显式状态	Retry / Replan / Verify / Escalate

Evaluation Metrics： Task Success Rate; Exception Recovery Rate; Policy Violation Rate; Role Routing Accuracy; Approval Validity Rate; Human Intervention Rate; Average Resolution Time; Tool Call Efficiency。应分别统计 AI 提出的越权建议和系统实际执行的越权操作。

Fault Injection： API 超时、重复/乱序事件、文件冲突、过期审批、外部执行结果未知及未解除 Hold 时的提货请求。

MVP Scenarios： (1) Missing Document; (2) Document Mismatch; (3) Customs Hold Coordination。建议先在模拟清关环境中开展评估，再考虑真实系统集成。

RQ1： Can an autonomous agent state machine improve task success compared with rule-based clearance workflows?

RQ2： How effectively does the proposed architecture recover from document, tool, and regulatory exceptions?

RQ3： Can explicit policy guards and verification reduce unauthorized actions while maintaining automation efficiency?

RQ4： Can role-aware human-in-the-loop routing improve assignment accuracy and prevent inappropriate approvals across customs-clearance stakeholders?

16 Implementation Requirements for HLD-to-LLD Transition

- **Action policy catalog:** classify every tool action by mode (AI, HYBRID, HUMAN/AUTHORITY), risk, required evidence and authorized executor.
- **RBAC + shipment scope:** define which individual accounts can review a given shipment and which role combinations or approval quorums are required.
- **Approval lifecycle:** requested → assigned → approved/rejected/expired; bind approvals to versioned payloads and revalidate guards before acting.
- **Event provenance:** keep internal human decisions separate from government/authorized official release messages.
- **Test matrix:** wrong-role approval, expired approval, modified draft after approval, two contradictory human answers, external release message missing, and admin attempting a broker-only action.

17 V3.0 Data-Aware Low-Level Design

17.1 Data Contract for Every Agent Node

Each node must define: **required inputs, typed fields, originating system, freshness/authority, evidence IDs, transformation logic, guard conditions, responsible human, output schema, and recovery route**. Agent states describe *execution*, not official customs dispositions. Source data, OCR extracts, AI hypotheses, human-reviewed findings, and official regulator outcomes must be explicitly distinguished.

17.2 Source Priority and Evidence Integrity

For conflicting values, the system *does not* apply a universal "latest document wins" rule. Instead it stores both observations, compares document purpose and version, identifies the responsible authority for that field, and escalates unresolved conflicts. A unique `evidence_id` links every claim and transition to source records. Confidence from OCR is extraction confidence, not a declaration of correctness.

17.3 Customs Data Dictionary

The dictionary below is an implementable *research schema*, not a claim that ACE publishes identical field names. The supplementary CSV includes all fields, constraints, consumers and owners.

表 8: Field-level data dictionary (research model)

Category	Field	Type	Source / Primary Check	Owner
D1	`shipment_id`	string	TMS; PK	R1
	`transport_mode`	enum	TMS; sea/air/land	R1
	`bill_of_lading`	string	Carrier; when applicable	R1/R4
	`air_waybill`	string	Carrier; when applicable	R1/R4
	`container_number`	string	Carrier; conditional	R1
	`port_code`	string	Carrier/TMS; required	R1
	`arrival_time_utc`	datetime	Carrier/terminal; optional pre-arrival	R1
	`carrier_reference`	string	Carrier; required	R1
D2	`document_id`	string	Document store; required	R3
	`document_type`	enum	Supplier/ERP; required	R3
	`document_version`	integer	Document store; positive	R3
	`invoice_number`	string	Commercial invoice; if invoiced	R3
	`seller_id`	string	Invoice/ERP; required	R3/R5
	`buyer_id`	string	Invoice/ERP; required	R3/R5
	`line_sku`	string	Invoice / packing list; per line	R3
	`line_quantity`	decimal	Invoice/packing/PO; positive	R2/R3
	`unit_price`	decimal	Invoice; nonnegative	R3/R5

Category	Field	Type	Source / Primary Check	Owner
	currency_code	ISO4217	Invoice; required	R3/R5
	document_hash	sha256	Document store; required	R7
D3	item_id	string	Product master; required	R5
	product_description	text	Supplier/Product master; required	R3/R5
	material_composition	text	Technical datasheet; conditional	R3
	product_function	text	Technical datasheet; conditional	R3
	country_of_origin	ISO country	Supplier declaration; required as applicable	R3/R5
	hts_candidate	string	AI + current USITC HTS; advisory only	R4/R5
	hts_reference_date	date	USITC HTS; versioned	R4/R5
	manufacturer_id	string	Supplier; conditional	R3/R5
D4	entry_number	string	Authorized filer; when issued	R4
	entry_type	string	Authorized filer; when available	R4
	filing_transaction_id	string	Broker integration; required for action	R4
	filing_status	enum	Broker/authorized status; source authenticated	R4
	rejection_code	string	Authorized message; when rejected	R4
	cbp_hold_status	enum	Trusted CBP/broker message; when applicable	R4/R6
	cbp_release_evidence_id	string	Trusted official channel; when released	R6
	official_message_timestamp	datetime	Authorized feed; required	R4/R6
D5	pga_agency	string	Regulatory rules/filing; conditional	R5
	pga_line_id	string	Broker/agency message; conditional	R4/R5
	pga_disposition	enum	Trusted agency/broker feed; conditional	R6
	permit_id	string	Importer / agency; conditional	R5
	product_code	string	Product master/agency rules; conditional	R5
	prior_notice_reference	string	FDA/authorized filer; food as applicable	R4/R5

Category	Field	Type	Source / Primary Check	Owner
	exam_instruction_id	string	Trusted authority notice; conditional	R4/R6
D6	declared_customs_value	decimal	Invoice + importer record; conditional	R5
	duty_estimate	decimal	Rules + broker; advisory	R4/R9
	duty_assessed	decimal	Authorized records; if available	R4/R9
	bond_reference	string	Importer/broker; conditional	R5/R9
	payment_status	enum	Finance/authorized feed; conditional	R9
	fee_reference	string	Finance/broker; conditional	R9
D7	task_id	uuid	Agent runtime; PK	R7
	agent_state	enum	State engine; required	R7
	action_type	enum	Tool registry; required	R7
	action_version	integer	Action store; monotonic	R7
	event_id	uuid	Event store; unique	R7
	evidence_id	uuid	Evidence store; required for claims	R7
	source_trust_level	enum	Adapter registry; required	R7
	approval_id	uuid	Approval manager; conditional	R5
	approval_status	enum	Approval manager; conditional	R5
	idempotency_key	string	Action engine; write actions	R7
	retry_budget	integer	Recovery policy; bounded	R7

Schema safeguards: UTC timestamps, currency codes, per-entry and per-line identifiers, document hashes, schema versioning, row-level organization access, action-version binding, and provenance metadata are mandatory wherever relevant.

17.4 Agent Node-Level Data Contracts

表 9: State–Data–Role–Output specification

Node	Required typed input	Output / Verifying guard	Escalation actor
INITIALIZE	task ID, shipment ID, importer ID, goal ID	versioned task; all IDs exist	R1
OBSERVE	source events, document IDs, timestamps, authority level	immutable observations; trusted source	R1 / R4
REASON	conflicting facts, evidence links, applicable rules	issue hypotheses, uncertainty flags, cited evidence	R4 / R5

Node	Required typed input	Output / Verifying guard	Escalation actor
COLLECT_DATA	missing field IDs, source owner, due date	verifiable data request; proper recipient	R2 / R3 / R5
PLAN	issue category, tools, data constraints, budgets	typed plan steps + expected outcome	R5 for high-risk
POLICY_CHECK	action type, RBAC, rule version, approval version	ALLOW / DENY / HUMAN; hard-guard result	role-specific reviewer
ACT	validated tool request, scoped credential, idempotency key	tool receipt and external transaction ref	R4 for filing
VERIFY	expected evidence, observed receipts, official events	verified step or remaining work; source checked	R4 / R5
WAIT_EVENT	pending event filter, deadline, correlation key	validated resumption event / timeout	assigned owner
RETRY	error class, retry budget, external status	retry only if reconciled + idempotent	R7 on unsafe retry
REPLAN	previous plan version, failure, side effects	new plan; re-check all guards	R4 / R5 if necessary
RECOVER	checkpoint, event log, transaction IDs	classified recovery choice + record	R7 or R4
ESCALATE	issue, evidence, role mapping, SLA	signed scoped approval or rejection	resolved role
COMPLETE	per-goal official / verified evidence set	clearance-ready task proof (not release issuance)	R4 / R5 as policy
FAILED	failure reason, events, unresolved side effects	retained record and human ownership	R1 / R4 / R7

17.5 Evidence-Based Completion Guard

Define `CUSTOMS_CLEARANCE_VERIFIED` as *all applicable* CBP, PGA and task-specific clearance requirements having trusted evidence. An unresolved Hold, pending applicable PGA line, rejected entry, unknown filing outcome, or stale approval must block completion. “Not applicable” is a separate enumerated value and cannot silently equal “cleared.” This research completion predicate must be validated against the shipment’s actual legal context by the authorized parties.

17.6 Human Data Ownership and Approval Boundaries

Role	Supplies / validates data	Can decide / cannot decide
Supplier (R3)	Invoice, packing list, specs, origin assertions	Corrects source documents; cannot issue CBP release
Warehouse (R2)	Inventory counts, photos, inspection evidence	Confirms physical facts; cannot approve customs filing
Broker (R4)	Entry, rejection, trusted customs messages	Performs authorized filing work; cannot invent CBP decision
Importer (R5)	Value, origin basis, importer facts, classification evidence	Responsible business approvals; cannot override CBP
Authority (R6)	Official hold, examination, PGA / release dispositions	Owns regulatory outcomes; not an internal AI assignee
Finance (R9)	Payment and bond evidence	Financial approvals; not customs release
Admin (R7)	Credentials, tool availability, system audit	Technical controls; not filing approval

18 V3.0 Customs Case Study Taxonomy

These cases are research-defined *scenarios*, not official CBP status labels. C15 here is re-scoped from post-release pickup to *partial customs / PGA line clearance*; transportation is excluded.

表 10: Eighteen clearance-focused benchmark cases

ID	Scenario	Data groups	Actor(s)	Expected evidence
C01	Missing commercial invoice	D2,D4	R3,R5 (HYBRID)	document present; filing held until evidence
C02	Quantity mismatch	D2,D3	R2,R3 (HYBRID)	reconciled counts, versioned file
C03	Invoice value conflict	D2,D6	R4,R5 (HYBRID)	human-reviewed valuation support
C04	HTS classification ambiguity	D3,D4	R4,R5 (HYBRID)	classification rationale approved
C05	Country-of-origin conflict	D2,D3	R3,R5 (HYBRID)	documented origin determination
C06	Trade remedy scope uncertainty	D3,D5,D6	R4,R5 (HYBRID)	broker/compliance review, never auto resolve
C07	Entry filing rejection	D4,D7	R4 (HYBRID)	authorized refiling and receipt
C08	Manifest or BL mismatch	D1,D4	R1,R4 (HYBRID)	trusted cross-reference evidence
C09	CBP customs hold	D4,D5	R4,R5,R6 (HYBRID+AUTHORITY)	official hold outcome

ID	Scenario	Data groups	Actor(s)	Expected evidence
C10	PGA line pending	D3,D5	R4,R5,R6 (HYBRID+AUTHORITY)	all applicable agency results confirmed
C11	Physical examination requested	D1,D4,D5	R4,R6 (AUTHORITY)	trusted inspection disposition
C12	Detention or refusal notice	D4,D5	R4,R5,R6 (AUTHORITY)	authorized next steps / case closure
C13	Duty estimate disagreement	D3,D6	R4,R9 (HYBRID)	finance/broker reconciliation
C14	Bond or payment issue	D4,D6	R4,R5,R9 (HYBRID)	authorized financial evidence
C15	Partial entry/line clearance	D4,D5	R4,R5 (HYBRID)	correct overall clearance gate
C16	Tool timeout unknown outcome	D4,D7	R7 (AI/HYBRID)	no duplicate external write
C17	Duplicate or out-of-order event	D4,D7	R7 (AI)	monotonic authoritative state
C18	Stale or conflicting approval	D7	R5,R7 (HYBRID)	new approval required after revision

18.1 Deep-Dive Case Contracts (Six Representative Cases)

Each deep-dive case includes the trigger, required fields, data owner, AI actions, mandatory authority checks, expected output and recovery transitions. All values in the examples are synthetic.

C02: Quantity mismatch

Trigger: Invoice quantity 500 vs packing-list quantity 480 for identical SKU.

Data: invoice.line_sku, invoice.line_quantity, packing.line_quantity, PO quantity, document version/hash, optional warehouse count

Owner: Supplier R3 supplies corrected document; Warehouse R2 verifies physical count if needed.

AI action: AI checks item matching and source versions; creates discrepancy report; requests confirmation.

Hard guard: Do not change filing value or asserted physical facts from OCR alone.

Expected path: OBSERVE > REASON > COLLECT_DATA > WAIT_EVENT > VERIFY; unresolved conflict > ESCALATE.

C04: Classification ambiguity

Trigger: Description is only ”electronic control module”; material/function missing.

Data: technical datasheet, composition, intended function, candidate HTS, effective date and source excerpt

Owner: R3 supplies specs; R4 and R5 review candidate classifications.

AI action: AI retrieves dated official HTS guidance and highlights unresolved properties.

Hard guard: Proposal is never treated as approved tariff classification; authorized reviewer must sign action version.

Expected path: REASON > COLLECT_DATA > PLAN > ESCALATE; after approval > POLICY_CHECK.

C07: Entry rejection

Trigger: Trusted broker feed reports rejected entry with code and filing reference.

Data: entry_number, filing_transaction_id, rejection_code, original payload hash, external submission status

Owner: R4 authorized broker/filer evaluates and performs lawful correction.

AI action: AI parses reason, locates candidate missing/invalid fields and drafts fix.

Hard guard: Unknown submission outcome requires inquiry before retry; no duplicate filing.

Expected path: OBSERVE > REASON > PLAN > POLICY_CHECK > ESCALATE > WAIT_EVENT > VERIFY.

C09: CBP hold / exam

Trigger: A trusted status feed indicates CBP Hold and possible document request.

Data: official status event, entry and line IDs, hold scope, time, demanded documents, notice evidence

Owner: R4 broker coordinates; R5 importer supplies facts; CBP R6 retains regulatory decision.

AI action: AI triages the notice, creates evidence-backed tasks, monitors deadlines and official updates.

Hard guard: AI must not clear the hold; stale/nonofficial release messages fail guard.

Expected path: OBSERVE > PLAN > COLLECT_DATA / ESCALATE > WAIT_EVENT > VERIFY.

C10: PGA line review

Trigger: Entry has multiple lines; one applicable PGA line is still pending.

Data: line IDs, agency, product codes, permit / prior notice as applicable, per-line disposition

Owner: Importer R5 and broker R4 supply/review; agency R6 controls disposition.

AI action: AI builds applicability matrix and merges agency status per entry line.

Hard guard: Completion requires every applicable line to have an acceptable official outcome.

Expected path: OBSERVE > REASON > WAIT_EVENT; incomplete line > COLLECT_DATA / ESCALATE.

C16: Unknown tool outcome

Trigger: Filing-oriented tool times out after request may have reached external system.

Data: idempotency key, transaction reference, request digest, response status, external inquiry results

Owner: R7 technical incident owner; R4 if legal filing impact.

AI action: AI checks transaction receipt and external status before any retry.

Hard guard: Never infer failure from timeout; block blind re-submission.

Expected path: ACT > RECOVER > WAIT_EVENT / ESCALATE > VERIFY > (RETRY only if safe).

19 V3.0 Events, API Contracts, and Failure Semantics

19.1 Normalized Event Envelope

```
{
  "event_id": "evt-001", "event_type": "CBP_HOLD_OBSERVED",
  "shipment_id": "SHP-1001", "entry_id": "ENTRY-01",
  "line_id": null, "occurred_at": "2026-10-15T16:00:00Z",
  "received_at": "2026-10-15T16:01:00Z",
  "source_type": "AUTHORIZED_BROKER_FEED",
  "source_message_id": "msg-881", "source_verified": true,
  "payload_schema_version": "1.0",
  "evidence_id": "evd-220", "correlation_id": "task-009"
}
```

Contract: adapters must preserve original payload and source identifiers; `event_id` must be unique; event time and ingestion time are separate; official results require provenance and verification appropriate to the source. Unverified email or LLM-produced text cannot change authority flags.

19.2 Tool Invocation Contract

```
{
  "action_id": "act-71", "action_version": 4,
  "tool_name": "request_supplier_document",
  "shipment_id": "SHP-1001", "requested_doc": "PACKING_LIST",
  "approval_id": null, "idempotency_key": "SHP-1001-act71-v4",
  "expected_evidence": "supplier_document_version",
  "timeout_seconds": 30
}
```

All write tools require policy approval, scoped authentication, schema validation, unique idempotency keys, and durable receipts. A timeout after an external request is treated as `UNKNOWN_OUTCOME` until reconciled. A filing tool remains mocked unless a compliant authorized integration is separately established.

19.3 Human Approval Contract

```
{
  "approval_id": "APR-88", "action_id": "ACT-78",
  "action_version": 2,
  "required_role": "LICENSED_CUSTOMS_BROKER",
  "shipment_id": "SHP-1001",
  "decision": "PENDING", "expires_at": "2026-10-16T20:00:00Z",
  "evidence_ids": ["evd-220", "evd-224"]
}
```

Approval must bind to organization, shipment, action, content hash and version. Any material edit invalidates the prior approval. Role routing and authorization are separate: identifying a nominal broker is not proof that the identity is authorized for the particular shipment.

19.4 Guarded Transition Pseudocode

```
handle_event(task_id, event):
  authenticate_and_validate(event)
  if event_id_seen(event.id): return ALREADY_PROCESSED
  with transaction_and_version_lock(task_id):
    task = load_task(task_id)
    append_raw_event(event)
    if is_authority_status(event):
      verify_source_correlation_and_scope(event)
      apply_authoritative_fact_if_newer(event)
    proposal = agent_propose(task, event)
    policy = check_guard_policy(proposal, task)
    if policy == DENY: persist(REPLAN); return
    if policy == APPROVAL: create_approval(task, proposal); return
    persist_transition(task, proposal.next_state)
    enqueue_action_via_outbox(proposal)
  return ACCEPTED
```

No LLM proposal may bypass the deterministic policy engine. Deduplication does not rely on timestamps alone; contradictory authoritative data is preserved and escalated for reconciliation.

20 V3.0 Storage Model, Metrics, and Test Cases

20.1 Core Database Entities

Entity	Key columns and constraints
`shipments`	shipment_id PK, importer_id, mode, business_status, row_version
`shipment_lines`	line_id PK, shipment_id FK, sku, origin, quantity, scope
`documents`	document_id PK, shipment_id FK, version, sha256, source, storage_uri
`field_observations`	observation_id PK, evidence_id FK, field, value, extractor, confidence
`entries`	entry_id PK, shipment_id FK, broker, external_reference, status
`agency_dispositions`	agency_id + line_id, official_source, disposition, evidence_id
`agent_tasks`	task_id PK, shipment_id FK, state, plan_version, row_version
`agent_actions`	action_id PK, task_id FK, version, tool, idempotency_key UNIQUE, outcome
`approvals`	approval_id PK, action_id FK, version, role, reviewer, content_hash, expiry
`events`	event_id PK UNIQUE, source_message_id, evidence_id, occurred_at, received_at
`recovery_attempts`	recovery_id PK, action_id FK, failure_class, budget, outcome
`audit_logs`	audit_id PK, actor, before_state, after_state, event_id, timestamp

20.2 Benchmark Protocol

18 case families with a proposed 20 variants each yield **360 synthetic tasks**; this is an experimental target, not an existing dataset. Each variant must include input records, expected allowed actions, expected reviewer, prohibited actions, authoritative event sequence, failure injections, and a ground-truth outcome. Split by scenario templates to avoid near-duplicate leakage between development and evaluation.

20.3 Evaluation Metrics and Acceptance Tests

- **Task success rate:** verified completed tasks / all eligible tasks; report by case type.
- **Role routing accuracy:** correctly routed escalations / total escalations requiring routing.
- **Unauthorized execution rate:** policy-violating executed actions / attempted actions (target zero in controlled tests).
- **Recovery success:** injected failures resolved without losing regulatory state integrity / injected recoverable failures.
- **Evidence completeness:** verified actions with full source and decision lineage / verified actions.
- **Time and cost:** end-to-end simulated resolution latency and per-case tool/LLM cost.

Critical negative tests: false official release assertion; CBP hold remains active; PGA line still pending; stale approval; document version change after signature; duplicate submission after timeout; delayed official message; unauthorized administrator attempts customs filing; conflicting source facts; absence of a required product attribute.

21 Implementation Plan and Limitations

MVP order: (1) data ingestion + evidence lineage; (2) C02 document mismatch; (3) C07 filing rejection simulator; (4) C09 hold and C10 PGA multi-line status; (5) approvals + unknown-outcome recovery; (6) fault-injection benchmark. Build read-only and mocked regulatory connectors first, never live filing by default.

Limitations: official procedural requirements vary by importer, goods, entry type, and agency; actual live ACE and broker access may not be available. The case taxonomy and benchmark are research proposals and require customs-practitioner review before treating the scenarios as fully representative. Regulatory text, data dictionaries and tariff schedules are versioned and must be refreshed for implementation.

22 Official Documentation and Provenance

1. U.S. Customs and Border Protection, Automated Commercial Environment (ACE): CBP ACE information.
2. CBP, Importing Into the United States: CBP importer guidance.
3. United States International Trade Commission, Harmonized Tariff Schedule: USITC HTS.
4. U.S. Food and Drug Administration, Entry Review: FDA entry review.
5. CBP, ACE Business Rules and Process Documentation: consult the current official specifications for status codes, versioning and EDI message structure. The identifiers used in this report are *internal simulation labels* rather than official code values.

23 Conclusion

Version 3.0 upgrades the conceptual autonomous customs agent from role-aware architecture to **field-level data lineage, typed node contracts, scenario-grounded approval routing, and testable evidence gates**. The objective is autonomous coordination under clear legal authority boundaries, not autonomous regulatory adjudication. All post-clearance pickup and transportation functions are excluded from this research phase.

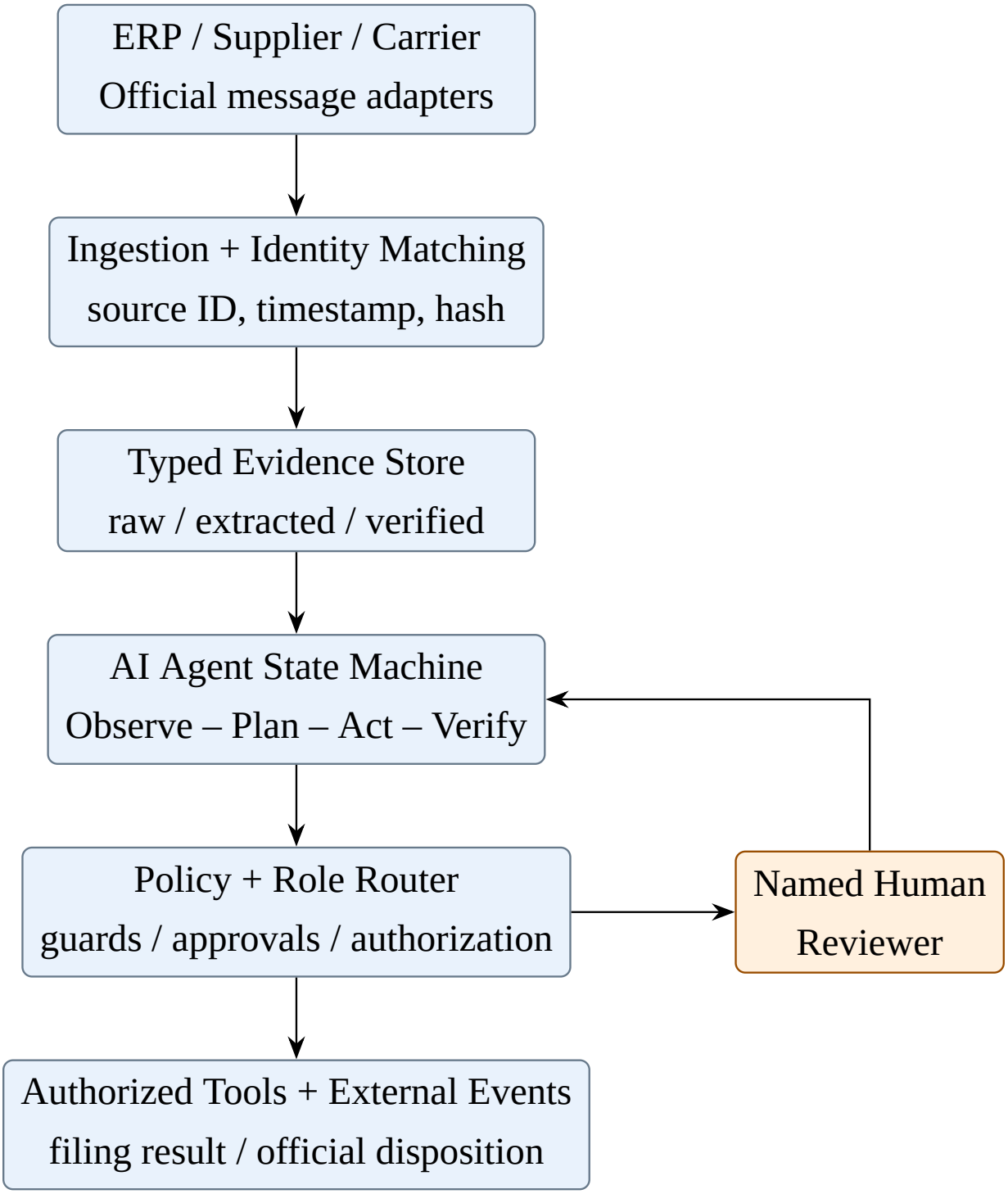


图 4: Customs data lineage and role-aware execution boundary. External authoritative outcomes are ingested as evidence, never created by an LLM.

A Complete Customs Data Dictionary (All CSV Fields)

This appendix reproduces the entire accompanying `Customs_Data_Dictionary_V3.csv`, including *category, field, type, source, constraint, usage, and owner role.* The earlier compact dictionary is retained in the main text for readability. Each entry below is a research-defined internal schema field, not necessarily an official ACE message element. Conditional requirements depend on shipment, commodity, entry type and regulator. All field names should be validated before production use.

表 11: Full customs data dictionary: all fields and metadata from the project CSV

Category	Field / Attribute	Type	Source	Constraint	Usage / Consumer	Owner
D1 Shipment / Transport	shipment_id	string	TMS	PK	join, provenance	R1
D1 Shipment / Transport	transport_mode	enum	TMS	sea/air/land	applicability	R1
D1 Shipment / Transport	bill_of_lading	string	Carrier	when applicable	manifest match	R1/R4
D1 Shipment / Transport	air_waybill	string	Carrier	when applicable	manifest match	R1/R4
D1 Shipment / Transport	container_number	string	Carrier	conditional	cargo identity	R1
D1 Shipment / Transport	port_code	string	Carrier/TMS	required	port validation	R1
D1 Shipment / Transport	arrival_time_utc	datetime	Carrier/terminal	optional pre-arrival	timeline	R1
D1 Shipment / Transport	carrier_reference	string	Carrier	required	join	R1
D2 Commercial Documents	document_id	string	Document store	required	provenance	R3
D2 Commercial Documents	document_type	enum	Supplier/ERP	required	document completeness	R3
D2 Commercial Documents	document_version	integer	Document store	positive	approval binding	R3
D2 Commercial Documents	invoice_number	string	Commercial invoice	if invoiced	cross-reference	R3
D2 Commercial Documents	seller_id	string	Invoice/ERP	required	party validation	R3/R5
D2 Commercial Documents	buyer_id	string	Invoice/ERP	required	party validation	R3/R5
D2 Commercial Documents	line_sku	string	Invoice / packing list	per line	line match	R3
D2 Commercial Documents	line_quantity	decimal	Invoice/packing/PO	positive	mismatch detection	R2/R3
D2 Commercial Documents	unit_price	decimal	Invoice	nonnegative	valuation research	R3/R5
D2 Commercial Documents	currency_code	ISO4217	Invoice	required	currency agreement	R3/R5
D2 Commercial Documents	document_hash	sha256	Document store	required	integrity	R7
D3 Product / Classification	item_id	string	Product master	required	line linking	R5
D3 Product / Classification	product_description	text	Supplier/Product master	required	HTS context	R3/R5
D3 Product / Classification	material_composition	text	Technical datasheet	conditional	classification evidence	R3
D3 Product / Classification	product_function	text	Technical datasheet	conditional	classification evidence	R3

Continued on next page

Complete Customs Data Dictionary (continued)

Category	Field / Attribute	Type	Source	Constraint	Usage / Consumer	Owner
D3 Product / Classification	country_of_origin	ISO country	Supplier declaration	required as applicable	origin check	R3/R5
D3 Product / Classification	hts_candidate	string	AI + current USITC HTS	advisory only	classification proposal	R4/R5
D3 Product / Classification	hts_reference_date	date	USITC HTS	versioned	rule freshness	R4/R5
D3 Product / Classification	manufacturer_id	string	Supplier	conditional	origin/PGA	R3/R5
D4 Customs Entry / Authority	entry_number	string	Authorized filer	when issued	entry correlation	R4
D4 Customs Entry / Authority	entry_type	string	Authorized filer	when available	workflow mapping	R4
D4 Customs Entry / Authority	filing_transaction_id	string	Broker integration	required for action	idempotency	R4
D4 Customs Entry / Authority	filing_status	enum	Broker/authorized status	source authenticated	rejection/acceptance	R4
D4 Customs Entry / Authority	rejection_code	string	Authorized message	when rejected	repair planning	R4
D4 Customs Entry / Authority	cbp_hold_status	enum	Trusted CBP/broker message	when applicable	hold gate	R4/R6
D4 Customs Entry / Authority	cbp_release_evidence_id	string	Trusted official channel	when released	release verification	R6
D4 Customs Entry / Authority	official_message_timestamp	datetime	Authorized feed	required	ordering	R4/R6
D5 Regulatory / PGA	pga_agency	string	Regulatory rules/filing	conditional	applicability	R5
D5 Regulatory / PGA	pga_line_id	string	Broker/agency message	conditional	line-level correlation	R4/R5
D5 Regulatory / PGA	pga_disposition	enum	Trusted agency/broker feed	conditional	release gate	R6
D5 Regulatory / PGA	permit_id	string	Importer / agency	conditional	document review	R5
D5 Regulatory / PGA	product_code	string	Product master/agency rules	conditional	PGA review	R5

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Complete Customs Data Dictionary (continued)

Category	Field / Attribute	Type	Source	Constraint	Usage / Consumer	Owner
D5 Regulatory / PGA	prior_notice_reference	string	FDA/authorized filer	food as applicable	food filing evidence	R4/R5
D5 Regulatory / PGA	exam_instruction_id	string	Trusted authority notice	conditional	exam coordination	R4/R6
D6 Finance / Bond	declared_customs_value	decimal	Invoice + importer record	conditional	valuation consistency	R5
D6 Finance / Bond	duty_estimate	decimal	Rules + broker	advisory	duty reconciliation	R4/R9
D6 Finance / Bond	duty_assessed	decimal	Authorized records	if available	comparison	R4/R9
D6 Finance / Bond	bond_reference	string	Importer/broker	conditional	bond checks	R5/R9
D6 Finance / Bond	payment_status	enum	Finance/authorized feed	conditional	payment evidence	R9
D6 Finance / Bond	fee_reference	string	Finance/broker	conditional	amount tracing	R9
D7 Agent / Audit	task_id	uuid	Agent runtime	PK	execution context	R7
D7 Agent / Audit	agent_state	enum	State engine	required	state transition	R7
D7 Agent / Audit	action_type	enum	Tool registry	required	policy	R7
D7 Agent / Audit	action_version	integer	Action store	monotonic	approval binding	R7
D7 Agent / Audit	event_id	uuid	Event store	unique	deduplication	R7
D7 Agent / Audit	evidence_id	uuid	Evidence store	required for claims	traceability	R7
D7 Agent / Audit	source_trust_level	enum	Adapter registry	required	authority priority	R7
D7 Agent / Audit	approval_id	uuid	Approval manager	conditional	human control	R5
D7 Agent / Audit	approval_status	enum	Approval manager	conditional	execution gate	R5
D7 Agent / Audit	idempotency_key	string	Action engine	write actions	safe retry	R7
D7 Agent / Audit	retry_budget	integer	Recovery policy	bounded	resilience	R7

B Complete Customs Case Studies Catalog

The following table reproduces the 18-case CSV catalog in full, including *required data groups, participating roles, execution mode, proposed agent actions, and success evidence*. Cases are synthetic test scenarios and their described workflow is not a substitute for advice from the importer, licensed customs broker, or competent authority. Only customs clearance and its regulatory completion evidence are in scope.

表 12: Complete customs case-study catalog: all fields from the project CSV

Case	Scenario	Required data	Role(s)	Execution mode	Proposed AI agent actions	Success / completion evidence
C01	Missing commercial invoice	D2,D4	R3,R5	HYBRID	request document; validate signature/source	document present; filing held until evidence
C02	Quantity mismatch	D2,D3	R2,R3	HYBRID	line-level compare; inventory confirmation	reconciled counts, versioned file
C03	Invoice value conflict	D2,D6	R4,R5	HYBRID	detect unit-price/value differences	human-reviewed valuation support
C04	HTS classification ambiguity	D3,D4	R4,R5	HYBRID	retrieve current HTS; explain alternatives	classification rationale approved
C05	Country-of-origin conflict	D2,D3	R3,R5	HYBRID	compare origin assertions	documented origin determination
C06	Trade remedy scope uncertainty	D3,D5,D6	R4,R5	HYBRID	flag AD/CVD/scope research	broker/compliance review, never auto resolve
C07	Entry filing rejection	D4,D7	R4	HYBRID	parse rejection; propose corrected draft	authorized refiling and receipt
C08	Manifest or BL mismatch	D1,D4	R1,R4	HYBRID	match references; route discrepancy	trusted cross-reference evidence
C09	CBP customs hold	D4,D5	R4,R5,R6	HYBRID+ AUTHORITY	track notice and authorized response	official hold outcome
C10	PGA line pending	D3,D5	R4,R5,R6	HYBRID+ AUTHORITY	verify line-level dispositions	all applicable agency results confirmed
C11	Physical examination requested	D1,D4,D5	R4,R6	AUTHORITY	coordinate required evidence; wait	trusted inspection disposition
C12	Detention or refusal notice	D4,D5	R4,R5,R6	AUTHORITY	extract deadlines and route legal review	authorized next steps / case closure
C13	Duty estimate disagreement	D3,D6	R4,R9	HYBRID	compare declared basis and current rules	finance/broker reconciliation
C14	Bond or payment issue	D4,D6	R4,R5,R9	HYBRID	detect status and evidence gaps	authorized financial evidence
C15	Partial entry/line clearance	D4,D5	R4,R5	HYBRID	aggregate line-level dispositions	correct overall clearance gate
C16	Tool timeout unknown outcome	D4,D7	R7	AI/HYBRID	reconcile status before resend	no duplicate external write
C17	Duplicate or out-of-order event	D4,D7	R7	AI	deduplicate/version and sequence	monotonic authoritative state
C18	Stale or conflicting approval	D7	R5,R7	HYBRID	bind approval to action version	new approval required after revision

C Catalog Traceability and Review Notes

The data dictionary provides internal field specifications; the case catalog maps scenarios to data groups (D1–D7), stakeholders and expected evidence. The detailed node contracts and six deep-dive case specifications in the main body describe how those data elements enter state transitions, guards and human-in-the-loop reviews. For live implementation, a customs specialist must review regulatory applicability and authoritative status messages. The tables above are synchronized with the companion CSV versions included in this research package.